



XYRA AI Studio

Private AI for EPCs

P&ID Intelligence

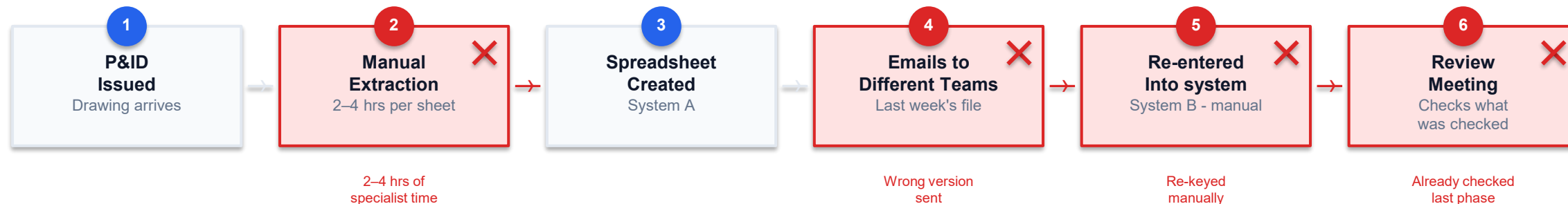
Piping MTO

Centralized Database

On-Premise

This is how engineering data moves today.

From the moment a P&ID is issued to the moment a deliverable is produced.



The same instrument tag is entered 4-6 times across a single project.
Discipline-to-discipline data transfer happens through disconnected files - no traceability, no version control.



"We'll use ChatGPT."

Doesn't know ISA-5.1 ✗

Can't reliably read P&IDs ✗

Client drawings can't go to a public AI ✗

30–50%

of your engineering budget
goes to clerical work.

Not engineering.

Where it goes

Breakdown of wasted engineering capacity on a typical EPC project:

Manual take-off

2–4 hrs per P&ID drawing, repeated across hundreds of drawings

Data re-entry

Same instrument tag entered 4–6× across systems by different teams

Discipline handoffs

Emailed spreadsheets, wrong versions, no audit trail between teams

Review & re-checking

Review phases repeat checks already done - no shared engineering record

There is a better way.

XYRA AI Studio

Private AI for EPC Drawing Deliverables

01

InstruMap

Extract instrument tags, classify with ISA-5.1, map to lines, generate Index / IO List / Verification Log.

02

Piping MTO

User-built component library + computer-vision detection across all drawings. EPC-style Excel output.

03

Centralized DB

PostgreSQL-backed engineering records - instruments, loops, piping, IO - shared across all disciplines.

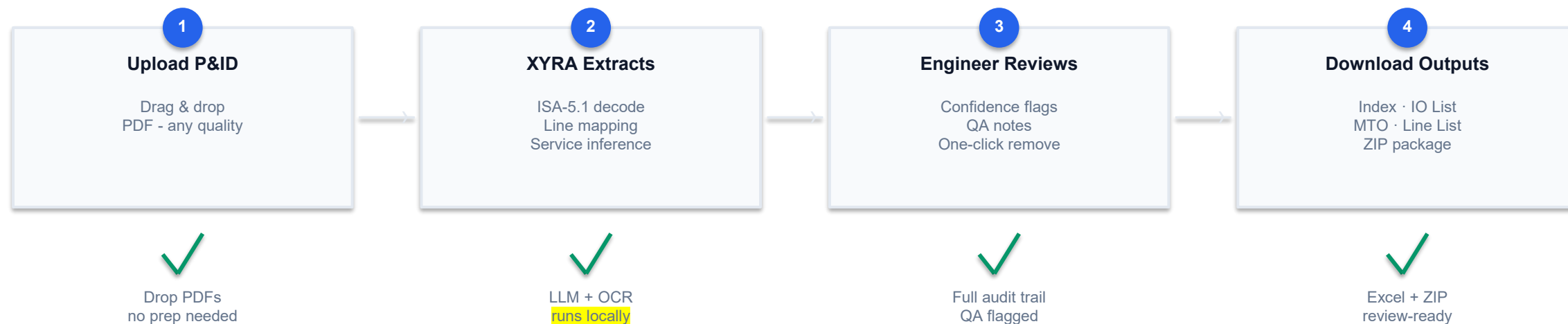
04

Custom AI Solutions

Build custom engineering applications, automate project-specific workflows, fine-tune private AI models, and train systems using client standards, specifications, and historical project data.

This is how it works with XYRA.

One upload. Every tool. One structured output to all disciplines.



BEFORE XYRA

Manual take-off → 5 systems re-entered → email → wrong version → re-check

WITH XYRA

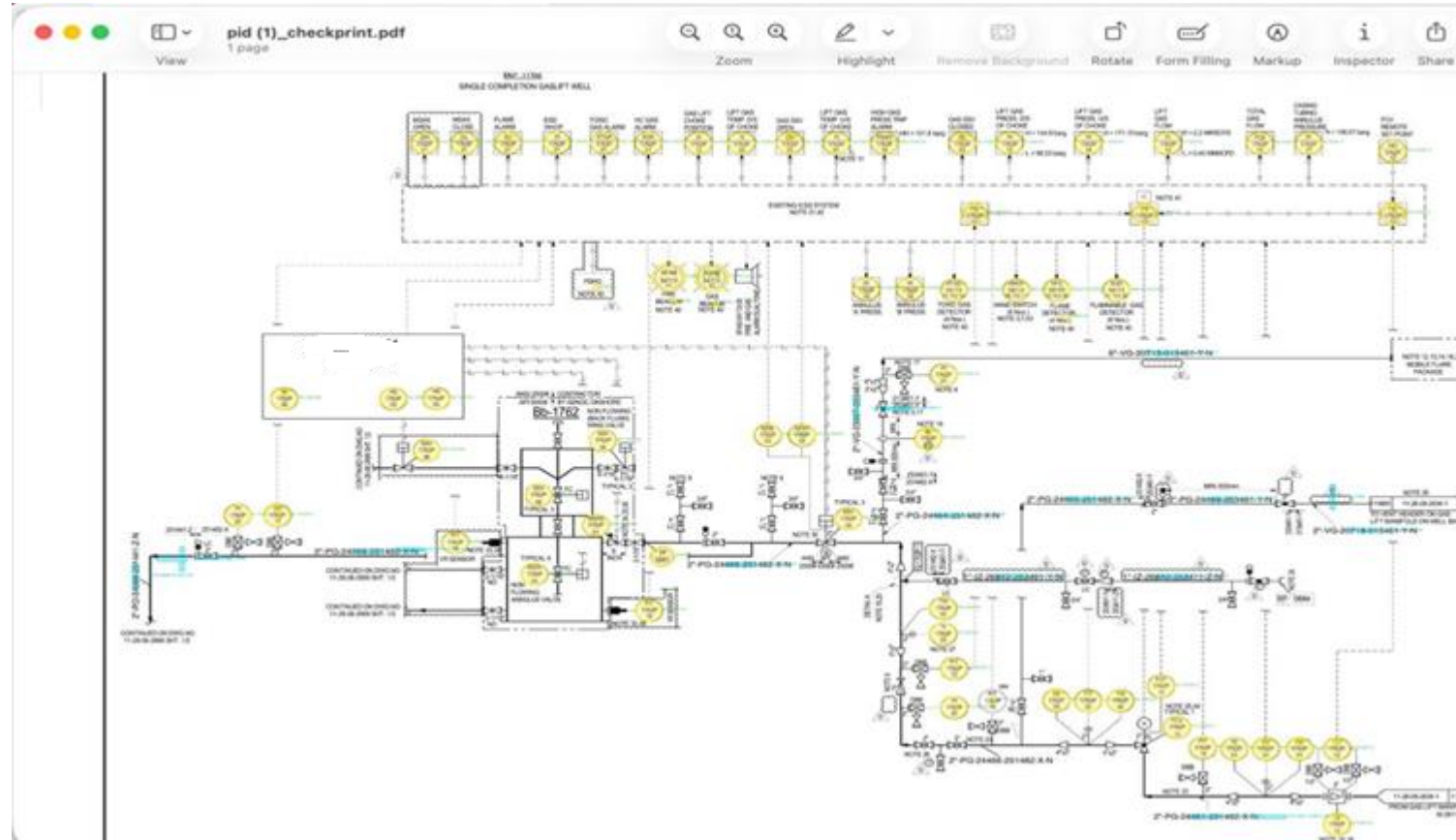
Upload → XYRA reads → engineer reviews → structured outputs to all disciplines

8 / 8 services live · 4 custom AI engines · 18.7 GB models loaded · 100% health · 0 queue



XYRA reads the drawing and prepares check print

Every instrument tag identified, classified, and linked to its pipeline number.



Every instrument.

Instrument Index - tags · loops · ISA type · service description · IO type · P&ID reference · QA review flags

Instrument Index

A2	fx	FCV-1762P-12											
	A	B	C	D	E	F	G	H	I	J	K	L	M
	Tag No.	Loop No.	Instrument Description	Instrument Service	Service Confidence	Service Basis	Instrument Ty	Suffi	P&ID No.	Line / Equip. Tag	Control System	IO Type	Signal Type
2	FCV-1762P-12	F-1762P-12	Flow Control Valve	Chemical Injection line valve control	Medium	connected line 1-IZ-2684	FCV	12	pid (1).pdf	1-IZ-26840-253411-Z-N	DCS	AO	4-20mA
3	FE-1762P-12	F-1762P-12	Flow Element	Flow element	Low	passive tag type only	FE	12	pid (1).pdf		DCS	Soft Link	
4	FI-1762P-12	F-1762P-12	Flow Indicator	VG service line flow indication	Medium	connected line 6-VG-207	FI	12	pid (1).pdf	6-VG-20715-013461-Y-N	DCS	Soft Link	
5	FIC-1762P-12	F-1762P-12	Flow Indicating Controller	Process flow control	Low	tag type only	FIC	12	pid (1).pdf		DCS	Soft Link	
6	FIT-1762P-12	F-1762P-12	Flow Indicating Transmitter	VG service line flow	Medium	connected line 2-VG-207	FIT	12	pid (1).pdf	2-VG-20718-013461-Y-N	DCS	AI	4-20mA + HART
7	FQI-1762P-12	F-1762P-12	Flow Totalizer Indicator	VG service line flow indication	Medium	connected line 6-VG-207	FQI	12	pid (1).pdf	6-VG-20715-013461-Y-N	DCS	Soft Link	
8	FZT-1762P-12	F-1762P-12	Flow Element Position Transmitter	Control valve position feedback	High	FZT position feedback	FZT	12	pid (1).pdf	1-IZ-26840-253411-Z-N	DCS	AI	4-20mA + HART
9	HS-1762P-03	H-1762P-03	Hand Switch	Operator hand switch	High	HS hand switch	HS	03	pid (1).pdf	2-PG-24468-251482-X-N1	DCS	DI	24VDC (Dry Contact)
10	HS-1762P-04	H-1762P-04	Hand Switch	Operator hand switch	High	HS hand switch	HS	04	pid (1).pdf	2-PG-24468-251482-X-N1	DCS	DI	24VDC (Dry Contact)
11	HS-1762P-06	H-1762P-06	Hand Switch	Operator hand switch	High	HS hand switch	HS	06	pid (1).pdf	2-PG-24468-251482-X-N1	DCS	DI	24VDC (Dry Contact)
12	HIC-1762P-12	H-1762P-12	Hand Indicating Controller	Process instrument control	Low	tag type only	HIC	12	pid (1).pdf		DCS	Soft Link	
13	MSAS-1762P-01	M-1762P-01	Moisture Switch Alarm Switch	PG service line instrument switch	Medium	connected line 2-PG-244	MSAS	01	pid (1).pdf	2-PG-24468-251482-X-N1	DCS	DI	24VDC (Dry Contact)
14	PI-1762P-12	P-1762P-12	Pressure Indicator	VG service line pressure indication	Medium	connected line 6-VG-207	PI	12	pid (1).pdf	6-VG-20715-013461-Y-N	DCS	Soft Link	
15	PIC-1762P-12	P-1762P-12	Pressure Indicating Controller	Process pressure control	Low	tag type only	PIC	12	pid (1).pdf		DCS	Soft Link	
16	PIT-1762P-12	P-1762P-12	Pressure Indicating Transmitter	Pressure near SSV-1762P-03	High	same line/loop valve cor	PIT	12	pid (1).pdf	2-PG-24464-251482-X-N	DCS	AI	4-20mA + HART
17	PY-1762P-12	P-1762P-12	Pressure Relay/Compute	Pressure signal conversion	Low	relay/converter tag type	PY	12	pid (1).pdf		DCS	Soft Link	
18	PI-1762P-14	P-1762P-14	Pressure Indicator	PG service line pressure indication	Medium	connected line 2-PG-244	PI	14	pid (1).pdf	2-PG-24461-251482-X-N	DCS	Soft Link	
19	PIT-1762P-14	P-1762P-14	Pressure Indicating Transmitter	PG service line pressure	Medium	connected line 2-PG-244	PIT	14	pid (1).pdf	2-PG-24461-251482-X-N	DCS	AI	4-20mA + HART
20	PIT-1762P-15	P-1762P-15	Pressure Indicating Transmitter	Chemical Injection line pressure	Medium	connected line 1-IZ-2684	PIT	15	pid (1).pdf	1-IZ-26842-253461-Y-N	DCS	AI	4-20mA + HART
21	PSAH-1762P-15	P-1762P-15	Pressure Switch Alarm High	Chemical Injection line pressure high switch	Medium	connected line 1-IZ-2684	PSAH	15	pid (1).pdf	1-IZ-26842-253461-Y-N	DCS	DI	24VDC (Dry Contact)
22	PI-1762P-17	P-1762P-17	Pressure Indicator	PG service line pressure indication	Medium	connected line 2-PG-244	PI	17	pid (1).pdf	2-PG-24468-251482-X-N1	DCS	Soft Link	
23	PIT-1762P-17	P-1762P-17	Pressure Indicating Transmitter	Pressure upstream of SSV-1762P-06	High	same line/loop valve cor	PIT	17	pid (1).pdf	2-PG-24468-251482-X-N1	DCS	AI	4-20mA + HART
24	PI-1762P-18	P-1762P-18	Pressure Indicator	Pressure indication downstream of SSV-1762P-06	High	same line/loop valve cor	PI	18	pid (1).pdf	2-VG-23007-253461-Y-N	DCS	Soft Link	
25	PIT-1762P-18	P-1762P-18	Pressure Indicating Transmitter	Pressure near SSV-1762P-06	High	same line/loop valve cor	PIT	18	pid (1).pdf	2-PG-24468-251482-X-N1	DCS	AI	4-20mA + HART
26	PI-1762P-19	P-1762P-19	Pressure Indicator	NOTE service line pressure indication	Medium	connected line 2-NOTE-2	PI	19	pid (1).pdf	2-NOTE-253461-Y	DCS	Soft Link	
27	PIT-1762P-19	P-1762P-19	Pressure Indicating Transmitter	Pressure downstream of SSV-1762P-04	High	same line/loop valve cor	PIT	19	pid (1).pdf	2-PG-24466-251482-X-N	DCS	AI	4-20mA + HART
28	PI-1762P-20	P-1762P-20	Pressure Gauge	Pressure indication near SSV-1762P-03	High	same line/loop valve cor	PI	20	pid (1).pdf	2-PG-24464-251482-X-N			
29	PI-1762P-21	P-1762P-21	Pressure Gauge	Pressure indication near SSV-1762P-03	High	same line/loop valve cor	PI	21	pid (1).pdf	2-PG-24464-251482-X-N			
30	PI-1762P-22	P-1762P-22	Pressure Gauge	Pressure indication upstream of SSV-1762P-06	High	same line/loop valve cor	PI	22	pid (1).pdf	2-PG-24468-251482-X-N18			
31	RO-1762P-03	R-1762P-03	Radiation Orifice	Radiation Orifice	Low	description fallback	RO	03	pid (1).pdf	2-NOTE-253461-Y	DCS	Soft Link	
32	SSV-1762P-03	S-1762P-03	Surface Safety Valve	Shutdown valve command	High	SSV shutdown valve	SSV	03	pid (1).pdf	2-PG-24464-251482-X-N	SIS/ESD	DO	24VDC
33	SZSH-1762P-03	S-1762P-03	Safety Position Switch (High/Open)	VG service line position high switch	Medium	connected line 2-VG-230	SZSH	03	pid (1).pdf	2-VG-23007-253461-Y-N	SIS/ESD	DI	24VDC (Dry Contact)
34	SZSL-1762P-03	S-1762P-03	Safety Position Switch (Low/Closed)	PG service line position low switch	Medium	connected line 2-PG-244	SZSL	03	pid (1).pdf	2-PG-24466-251482-X-N	SIS/ESD	DI	24VDC (Dry Contact)
35	SSSV-1762P-04	S-1762P-04	Subsurface Safety Valve	Shutdown valve command	High	SSSV shutdown valve	SSSV	04	pid (1).pdf	2-PG-24466-251482-X-N	SIS/ESD	DO	24VDC
36	SSV-1762P-05	S-1762P-05	Surface Safety Valve	Shutdown valve command	High	SSV shutdown valve	SSV	05	pid (1).pdf	2-VG-23007-253461-Y-N	SIS/ESD	DO	24VDC
37	SSV-1762P-06	S-1762P-06	Surface Safety Valve	Shutdown valve command	High	SSV shutdown valve	SSV	06	pid (1).pdf	2-PG-24468-251482-X-N1	SIS/ESD	DO	24VDC
38	SSV-1762P-08	S-1762P-08	Surface Safety Valve	Shutdown valve command	High	SSV shutdown valve	SSV	08	pid (1).pdf		SIS/ESD	DO	24VDC

Every line number.

Line List - pipe numbers · sizes · fluid codes · insulation spec · area codes · coordinates · P&ID source

AutoSave

Line List

HomeInsertDrawPage LayoutFormulasDataReviewViewAutomate

Paste

Cut

Copy

Format

Calibri (Body)11A[^]A_v

B I U

Wrap Text Merge & Centre

General

Conditional Formatting Format as Table Cell Styles

Insert Delete

D31

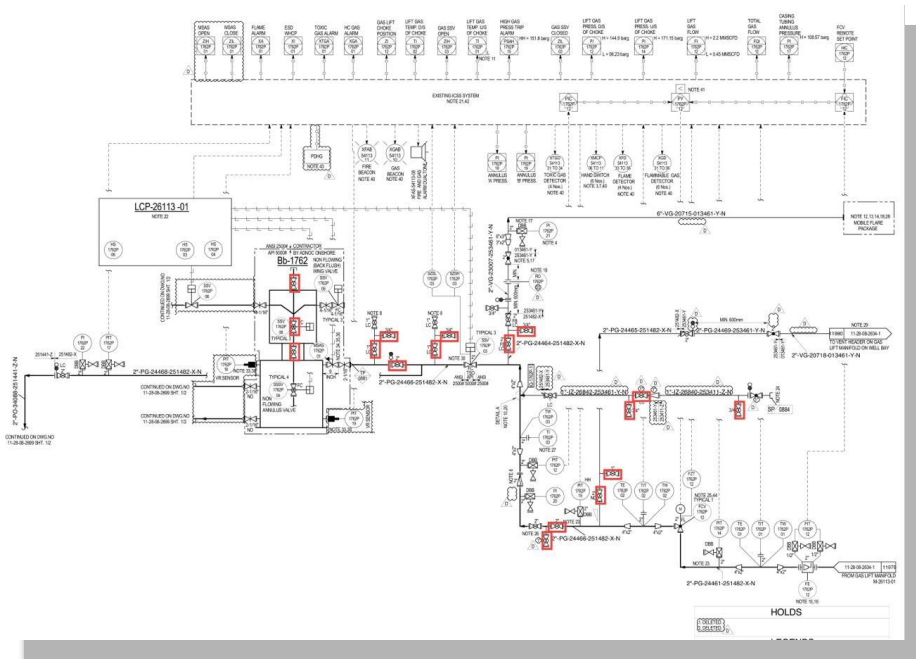
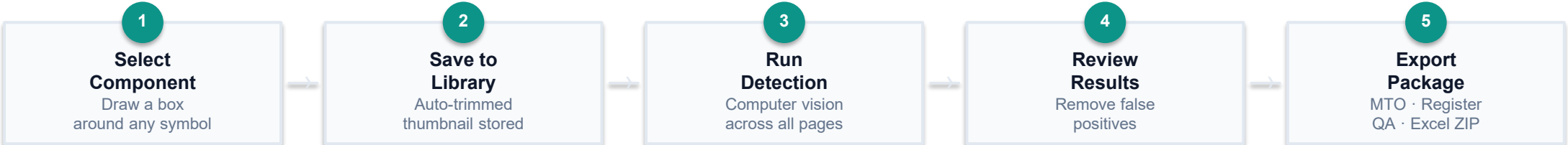
	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Line Number	Pipe Size	Size Unit	Fluid Code	Sequence	Area Code	Insulation	Spec	P&ID Filename	P&ID Pa	Coordinates			
2	2-PG-24466-251482-X-N	2 in	PG	24466	251482	X	N		pid (1).pdf	1	3975,3614			
3	2-PG-24465-251482-X-N	2 in	PG	24465	251482	X	N		pid (1).pdf	1	5839,3181			
4	2-VG-20718-013461-Y-N	2 in	VG	20718	013461	Y	N		pid (1).pdf	1	7410,3390			
5	2-PG-24469-253461-Y-N	2 in	PG	24469	253461	Y	N		pid (1).pdf	1	6616,3182			
6	2-PG-24461-251482-X-N	2 in	PG	24461	251482	X	N		pid (1).pdf	1	6535,5286			
7	2-PG-24464-251482-X-N	2 in	PG	24464	251482	X	N		pid (1).pdf	1	5119,3289			
8	2-VG-23007-253461-Y-N	2 in	VG	23007	253461	Y	N		pid (1).pdf	1	4617,2587			
9	6-VG-20715-013461-Y-N	6 in	VG	20715	013461	Y	N		pid (1).pdf	1	6289,2212			
10	2-PG-24468-251482-X-N	2 in	PG	24468	251482	X	N		pid (1).pdf	1	1929,3527			
11	2-PO-34088-251441-Z-N	2 in	PO	34088	251441	Z	N		pid (1).pdf	1	660,3690			
12	1-IZ-26842-253461-Y-N	1 in	IZ	26842	253461	Y	N		pid (1).pdf	1	5477,3695			
13	1-IZ-26840-253411-Z-N	1 in	IZ	26840	253411	Z	N		pid (1).pdf	1	6351,3694			
14	2-PO-34088-251441-Z-NDBB	2 in	PO	34088	251441	Z	NDBB		pid (1).pdf	1	916,3579			
15	2-PG-24461-251482-X-NFE	2 in	PG	24461	251482	X	NFE		pid (1).pdf	1	6895,5289			
16	2-D-11980	2 in	D	11980					pid (1).pdf	1	7231,3169			
17	2-PG-24468-251482-X-N18	2 in	PG	24468	251482	X	N18		pid (1).pdf	1	2173,3517			
18	2-NOTE-253461-Y	2 in	NOTE	253461	Y				pid (1).pdf	1	4830,2580			
19	2-PO-34088-251441-Z-NLC	2 in	PO	34088	251441	Z	NLC		pid (1).pdf	1	860,3582			
20														

www.xyra-ai.com

Confidential

Select once.

XYRA finds every instance - across every page, every rotation, with size extraction.



OUTPUT

	A	B	C	D
1	No.	Component	Image	Count
2	1	Ball_Valve_O		17
3				
4				
5		TOTAL		17
6				
7				
8				

17 components detected

1 drawing processed

1 component type

Beyond Products – Engineering Automation Services

Engineering Tool Automation

- Custom DLLs and add-ins for Hexagon SmartPlant tools
- SmartPlant Instrumentation (SPI) automation
- Smart 3D (S3D) workflow automation
- Smart P&ID data extraction and validation
- Custom engineering productivity tools

Engineering Data Management

- Engineering database migration and cleansing
- Bulk data loading and validation
- Legacy data digitization
- Intelligent document processing

AI-Powered Engineering Solutions

- Private/on-premise AI assistants
- Custom AI models trained on engineering data
- Drawing intelligence and knowledge extraction
- Engineering document search and Q&A

Two weeks. One answer.

A structured pilot gives your team a clear decision: time saved, output quality, and deployment fit.

WEEK 1

Configure & Load

- ✓ Deploy on client server or test VM
- ✓ Load project legend + title block context
- ✓ Configure instrument model + MTO library
- ✓ Run first extraction pass - calibrate output

WEEK 2

Validate & Compare

- ✓ Process 20–50 representative P&IDs
- ✓ Compare XYRA output vs manual workflow
- ✓ Review Instrument Index, IO List, MTO
- ✓ Identify gaps - tune detection settings

DECISION PACK

Delivered at Close

- ✓ Coverage summary + exception log
- ✓ Time-saving estimate vs manual baseline
- ✓ Deployment checklist for production
- ✓ Monthly license proposal



Ready to see XYRA on your drawings?

We deploy inside your network, run XYRA on your drawing package,
and deliver structured outputs your team evaluates against the manual baseline.

[Schedule a Pilot Run](#)